

A *Avatar* took four-and-a-half years to make and spans a period in the entertainment industry where a lot has changed. Those changes both affected *Avatar* and the film also affected those changes.

I became a cinema lover as a child after watching films like Ray Harryhausens's *The 7th Voyage of Sinbad* and *Jason and the Argonauts*. They were pure magic. I love science fiction and fantasy and later as I became a filmmaker, I worked in that genre and my first film was *The Terminator*, and then I did *Aliens* and then I became a practitioner of visual effects. For *The Abyss*, 22 years ago, we used the CGI or "computer generated imagery" to create the pseudopod. Though the film didn't do well, it had a magical effect on the audience and I realised the power of CG. I used it in *Terminator 2* in 1991 for the liquid man. The film was successful beyond our expectations and I realised that technology that enthralled audiences is something I needed to pursue. I started Digital Domain to pursue CG to raise art and technology. In 1995, I wrote *Avatar*, with its creatures, fantastic settings and blue people with golden eyes. We decided to wait for the industry to come up with the capability to make this movie possible.

Another thing that fascinated me in 1995 was 3D. I made a 3D film and it was screened in IMAX as there were no 3D theatres at that time. The cameras were the size of refrigerators. So around 2000, I started to work to build small-scale digital cameras based on HD technology with which I could shoot in 3D. I wanted to shoot with these cameras but we could show them only in IMAX. So we used 3D cameras to make documentary films over the next few years. After working with these cameras, I decided I would only make 3D digital films. Looking at the image on the screen coming from the 3D cameras was like looking through a window into reality. If I could get that power into my filmmaking, I believed I could do something extraordinary. The only problem was that there were no theatres to show such movies. So I thought, I will continue to work in documentaries and meanwhile will evangelise theatre owners to adopt this new technology. My vision was a digital cinema revolution and digital cinema as the way we watch movies. Though we could save over a billion dollars on prints a year by digitising prints, nobody wanted to move forward. My vision of 3D riding on digital cinema was not happening. Then things started to change and theatres began conversions into digital cinemas, but it was not changing fast enough for me. I realised it would happen only when I made an amazing 3D film. I began work on *Avatar* in May 2005. The first three years consisted of getting a team of some of the best sketch artists in the world to create the world of the Na'vi and another team to create the technology. I knew that the film's success or failure would

be based on the close-up of the CG characters. If they didn't have the heart and soul of real people, the film would not work. We decided to do performance capture, which is not photography but capturing what the actor does. The camera work and lighting are done separately without the artist.

It had been done before, but the standard wasn't as good. We created image-based facial performance capture by putting a small helmet with a tiny boom with a standard definition video camera on the actors' heads. It would shoot their faces and close-ups of how their lips and eyes moved and tongue interacted. This data was streamed into a computer model of the character and brings the model to life. I was concerned when the actors said they did not want to act with a helmet on their heads and a camera in their face. But after a first few sessions, they seemed to love it. They worked for more than a year and never lost interest and I figured out why. For performance capture, you have to give a performance just once. All the camera coverage is done later. Actors loved it as it was a pure form of acting. It was also a pure form of directing.

After those performances arrived, we began work. We made mistakes and we learnt from it. It was probably the most stimulating moment of my career because here I was directing for more than 25 years and I did not have all the answers. We were so far from the frontier that we had to figure it out ourselves. My primary tool as director was a virtual camera where you could see Sam Worthington and Zoe Saldana as if they were in the jungle of Pandora as the 10-ft tall blue characters. It was like a window into another world. For live action, we used the 3D fusion camera system. Shooting live action took about four months after which we went into post-production. We spent \$250 million and three years before we saw one shot that proved that what we set in motion was going to work. The shot was the close-up of Neytiri looking at Jake. Everybody

thinks she is going to shoot him with her bow but she decides not to. You can see that decision play out in her eyes. She was alive; she looked real. We saw the scene and realised this was going to work. But we had to do it with 2,600 other shots. We managed all this in 18 months, finishing 1,200 shots in the last five weeks. Our biggest problem was marketing because we didn't have a Tom Cruise or a Brad Pitt nor was it part of a franchise like *Spiderman*.

So we went back to the exhibitors. At the cine-expo in Amsterdam, we showed 25 minutes of the film in 3D. We got rave reviews from the exhibitors and press. The exhibitors began preparations to put in more screens before the film released. When the picture opened on December 18, it did well. Not a record weekend but we did \$75 million which was comparable to *Alice in Wonderland* which did \$115 million. But I wasn't concerned because I believed that if the film got to

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JAMES CAMERON